

ITA0612 – MACHINE LEARNING

CO1-AT2

CONCEPT MAPPING

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QUESTIONS:

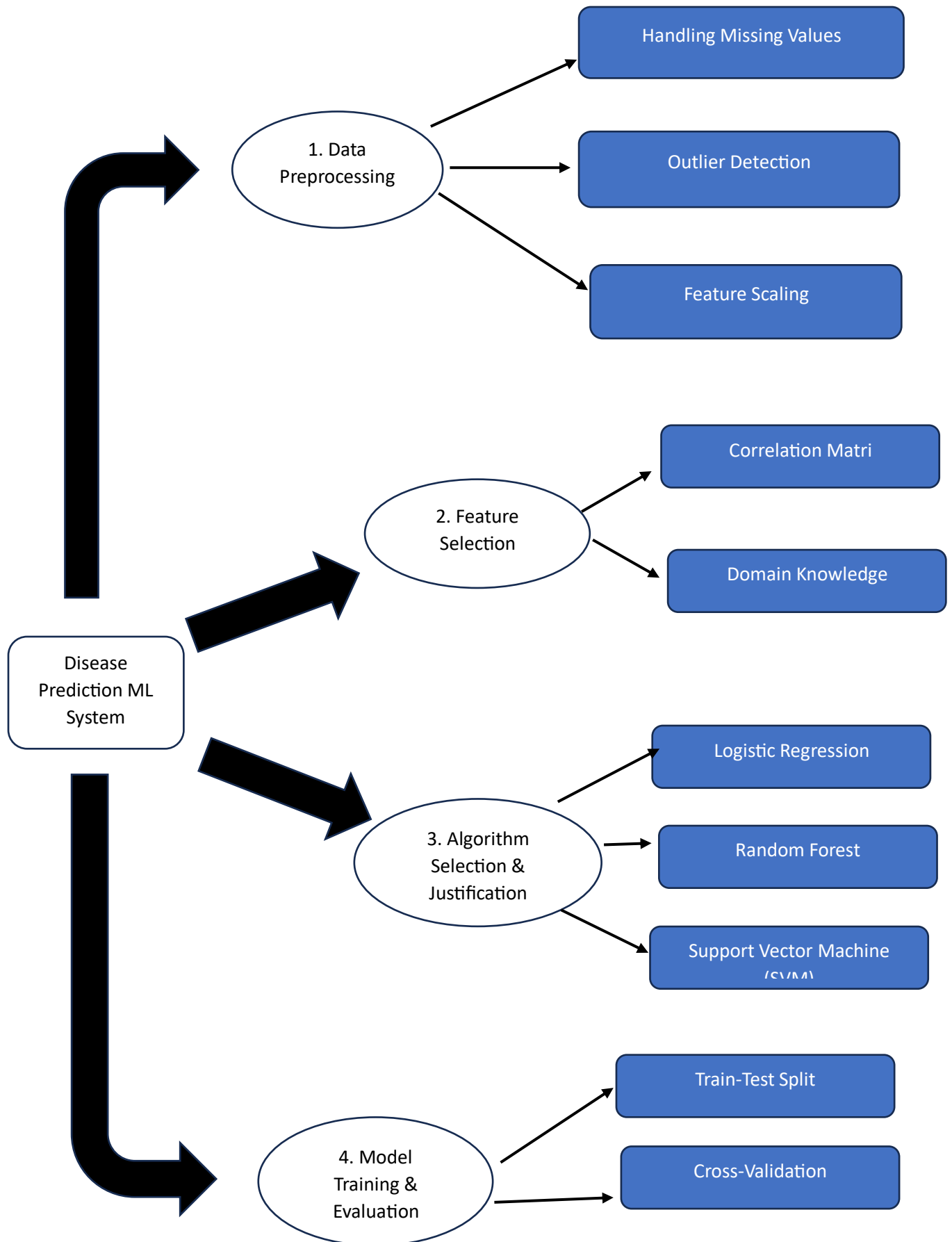
1. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors.

Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.

2. . In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.

Dataset:					
Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

1. machine learning system for health analysis to predict diseases



2. S ALGORITHM

Find-S Algorithm — Loan Approval
Concept map: central topic with color-coded branches for process, algorithm steps, evaluation, and output.

